DeepGuard

Documentation and Testing

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Document Scope: This document covers the complete testing plan for DeepGuard: all four testing levels with individual test cases and pass/fail criteria, all 50 fault simulation scenarios, every evaluation metric defined in plain English, the audit log field specification, and the long-term maintenance plan for keeping the system accurate over a multi-year mission.

1. Testing Philosophy

1.1 Why Testing Matters for an Autonomous Safety System

DeepGuard makes decisions that directly affect crew safety. A false alarm distracts the crew during a critical activity. A missed emergency can end the mission. Testing does not just check whether the system works. It checks whether the system fails safely, behaves consistently under stress, and matches what experienced aerospace engineers would decide in the same situation.

1.2 Four-Level Framework

Testing follows NASA's four-level framework from the Fault Management Handbook (NASA/TM-2012-217848). Each level builds on the previous one. No level is skipped. A failure at any level stops testing until the issue is resolved.

Level	Name	What Is Tested	Who Runs It
1	**Individual Tool Testing**	Each AI tool is tested alone, in isolation from the others, to confirm it meets its own specifications	AI developer (Robert B Thompson)
2	**Full System Testing**	All three tools are connected and tested together as one pipeline to confirm they work correctly as a unit	AI developer + independent reviewer

Level	Name	What Is Tested	Who Runs It
3	**Worst-Case Simulation**	The full system faces 50 real-world fault scenarios from NASA's Lessons Learned database, each run 10 times with added sensor noise	AI developer + independent reviewer
4	**Human Expert Review**	Aerospace engineering experts review 20 selected decision logs to confirm the system's choices match professional judgment	Faculty advisor + 5 engineering students

2. Level 1: Individual Tool Testing

2.1 Sensor Watcher Tests

Test ID	Test Name	How the Test Is Run	Pass Condition
SW-01	**Known fault detection**	500 sensor readings with known fault labels from NASA ISS data are fed to the Sensor Watcher one at a time	At least 435 of the 500 known faults are correctly flagged (87% detection rate)
SW-02	**False alarm rate**	5,000 sensor readings confirmed as normal are fed to the Sensor Watcher; count how many are incorrectly flagged as suspicious	No more than 250 of the 5,000 normal readings are flagged (5% false alarm ceiling)
SW-03	**Response speed**	10,000 readings are fed in sequence; the time from input to output is measured for every reading	Every single reading is scored in under 5 milliseconds with no exceptions
SW-04	**Cross-system correlation**	Inject a fault that only appears abnormal when two sensors from different systems are compared together simultaneously	Fault is correctly flagged within 3 sensor readings (0.3 seconds) of the anomalous pattern starting
SW-05	**Re-learning accuracy**	Run the 30-day model update cycle on simulated data with known sensor drift; then test detection accuracy again	Detection rate stays above 87% after re-learning on the drifted sensor data
SW-06	**Memory usage**	Run the Sensor Watcher continuously for 72 hours on simulated data; measure RAM usage every 10 minutes	RAM usage never exceeds 400 megabytes at any point during the full 72-hour run

2.2 Trend Forecaster Tests

Test ID	Test Name	How the Test Is Run	Pass Condition
TF-01	**Prediction accuracy on unseen data**	The forecaster trains on 80% of sensor history; the remaining 20% is used for predictions the model has never seen before	Average prediction error stays below 2% of each sensor's full reading range across all test readings
TF-02	**Worst-case single prediction**	From the unseen 20% test set, identify the single least accurate prediction made by the forecaster	Even the worst single prediction stays within 8% of the sensor's full reading range
TF-03	**Forecast speed**	Measure the time to produce a full 15-minute forecast for all sensors simultaneously	Complete forecast produced in under 50 milliseconds for all sensors at once

Test ID	Test Name	How the Test Is Run	Pass Condition
TF-04	**Accuracy after sensor drift**	Simulate 3 years of sensor aging in training data; re-train on the drifted data; measure prediction accuracy again	Average prediction error stays below 3% of sensor range on aged-sensor simulations
TF-05	**Early warning lead time**	Inject 20 scenarios where a sensor will cross a danger threshold; measure how many minutes advance warning the forecaster gives	At least 10 minutes of advance warning provided in 18 out of 20 scenarios

2.3 Alert and Action Engine Tests

Test ID	Test Name	How the Test Is Run	Pass Condition
AE-01	**Alert level accuracy**	200 scripted fault situations are presented, each with a suspicion score, forecast value, and spacecraft state; the engine assigns an alert level	Correct alert level assigned in all 200 scenarios with zero exceptions
AE-02	**Protocol selection accuracy**	From the same 200 scenarios, the engine selects a response protocol; the selection is compared to the expected answer	Correct protocol selected in all 200 scenarios with zero exceptions
AE-03	**Out-of-scope fault handling**	10 fault scenarios are presented that fall completely outside the pre-approved protocol library	All 10 out-of-scope scenarios result in a crew alert and no automated action being taken
AE-04	**Log entry completeness**	After running 50 scenarios, every log entry produced by the engine is reviewed for completeness	Every log entry contains all required fields: timestamp, sensor values, score, forecast, alert level, protocol ID, and checksum
AE-05	**Decision speed**	Measure the time from receiving inputs to completing the log entry across 1,000 consecutive scenarios	All 1,000 decisions completed in under 2 milliseconds each

2.4 Data Collection Layer Tests

Test ID	Test Name	How the Test Is Run	Pass Condition
DC-01	**Corrupted packet handling**	5% of input packets are deliberately corrupted with random bit errors before being fed to the layer	All corrupted packets are identified and discarded; no corrupted values reach the Sensor Watcher
DC-02	**Short sensor gap filling**	Individual sensors are silenced for 1 to 3 consecutive readings at random intervals during a 1-hour test run	All gaps filled using linear interpolation; filled values stay within 1% of the mathematically correct value
DC-03	**Long sensor gap flagging**	A sensor is silenced for more than 3 consecutive readings (more than 0.3 seconds)	Last-known-good value is held; a sensor-fault flag is raised within 1 reading of the gap starting
DC-04	**Normalization correctness**	100 sensor readings with known raw values are passed through the normalization step	All 100 normalized values fall within 0.001 of their mathematically correct 0-to-1 equivalents

3. Level 2: Full System Testing

Level 2 connects all four layers and tests them as one complete pipeline. The goal is to confirm that data flows correctly between layers, that end-to-end timing targets are met, and that the system handles conditions that individual layer tests cannot replicate.

Test ID	Test Name	How the Test Is Run	Pass Condition
SYS-01	**End-to-end response time**	10,000 sensor packets containing known faults are injected; time from packet arrival at Layer 1 to completed log entry in Layer 4 is measured for every packet	100% of fault events receive a completed log entry within 2 seconds of the sensor reading arriving; no exceptions
SYS-02	**Simultaneous multi-system faults**	Faults are injected in three different spacecraft systems at exactly the same moment; each must be detected and handled independently	All three faults are detected, classified at the correct alert level, and responded to independently; no fault suppresses detection of another
SYS-03	**Sensor dropout mid-run**	During a 4-hour continuous simulation, individual sensors are dropped at random for up to 30 seconds each; the system must keep running	System continues running throughout; no Emergency alerts generated because a sensor dropped out; all dropouts correctly logged
SYS-04	**72-hour continuous operation**	The full pipeline runs without interruption for 72 hours on simulated data, processing 10 readings per second from all sensors	Total RAM stays below 2 gigabytes; no crashes or unhandled errors; log file grows at the expected rate with no missing entries
SYS-05	**Alert escalation path**	A fault is injected that starts at Watch level and gradually worsens over 20 minutes until it reaches Emergency level	System correctly moves through Watch, Caution, Warning, then Emergency; each transition triggers the correct response
SYS-06	**Crew override during active response**	A Warning-level fault is triggered; while the Action Engine is executing the protocol, a crew override command is issued	The response protocol stops immediately upon receiving the override; override is logged; no further automated action taken until a new fault is detected

4. Level 3: Worst-Case Scenario Simulation

4.1 Simulation Setup

Fifty fault scenarios drawn from NASA's Lessons Learned Information System (llis.nasa.gov) are run 10 times each with a different random noise pattern added to the sensor data every run. This gives 500 total simulation runs and confirms the system behaves consistently even when sensor data is not clean.

Pass condition for Level 3: Zero Emergency-level faults missed across all 500 simulation runs. Detection rates for Watch, Caution, and Warning levels must each remain above 85%. No scenario produces a response at a higher severity than the actual fault warrants.

4.2 All 50 Fault Scenarios

ID	System	Fault Description	Starting Alert	Peak Alert
S-01	**Thermal**	Primary heater slowly drifting 15% above setpoint over 6 hours	Watch	Warning
S-02	**Thermal**	Radiator panel temperature dropping faster than coolant loss rate can explain	Caution	Warning
S-03	**Thermal**	Sudden temperature spike in electronics bay lasting under 2 minutes	Watch	Watch
S-04	**Thermal**	Thermal control fluid loop pressure declining 0.2 kPa per hour	Watch	Caution
S-05	**Thermal**	Two heating zones both rising simultaneously (correlated fault)	Caution	**Emergency**
S-06	**Power**	Single solar panel output drops to 55% of rated after micrometeorite hit	Caution	Warning
S-07	**Power**	Battery charge declining faster than normal discharge curve	Watch	Warning
S-08	**Power**	Main distribution bus voltage sagging 8% below nominal	Caution	Warning
S-09	**Power**	Unexpected circuit breaker trip on life support power feed	Watch	**Emergency**
S-10	**Power**	Solar panel output oscillating rapidly due to pointing instability	Watch	Caution
S-11	**Life Support**	CO2 partial pressure rising 0.05 kPa per hour above normal	Watch	Warning
S-12	**Life Support**	Cabin pressure declining 0.15 kPa per hour (slow leak)	Caution	**Emergency**
S-13	**Life Support**	Rapid cabin pressure drop of 2 kPa in under 5 minutes	Emergency	**Emergency**
S-14	**Life Support**	CO2 scrubber flow rate declining 20% below specification	Caution	Warning
S-15	**Life Support**	Oxygen partial pressure drifting below 19.5%	Watch	Warning
S-16	**Life Support**	Humidity rising above 70% for more than 2 hours	Watch	Caution
S-17	**Life Support**	Both primary and backup CO2 scrubbers showing degraded performance	Warning	**Emergency**
S-18	**Navigation**	Primary star tracker and gyroscope estimates disagreeing by 0.8 degrees	Watch	Caution
S-19	**Navigation**	Single gyroscope reporting drift rate 5x higher than the others	Caution	Warning
S-20	**Navigation**	Reaction wheel speed exceeding upper operational limit by 12%	Watch	Caution
S-21	**Navigation**	Attitude control system making unusually frequent corrections (oscillation)	Watch	Caution
S-22	**Navigation**	All three reaction wheels approaching saturation simultaneously	Warning	Warning
S-23	**Navigation**	Primary attitude sensor returning identical values for 30 seconds (freeze fault)	Watch	Warning

ID	System	Fault Description	Starting Alert	Peak Alert
S-24	**Propulsion**	Propellant tank pressure 3% below pre-burn nominal before a scheduled burn	Watch	Caution
S-25	**Propulsion**	Engine chamber pressure reading 8% off nominal during burn	Caution	Warning
S-26	**Propulsion**	Propellant flow meter showing lower flow than valve position implies	Watch	Warning
S-27	**Propulsion**	Thruster valve position sensor and command state disagree	Watch	Caution
S-28	**Propulsion**	Unexpected thrust detected during a non-burn period (stuck valve)	Warning	**Emergency**
S-29	**Propulsion**	Fuel and oxidizer flow rates diverging from the expected mixture ratio	Caution	Warning
S-30	**Communications**	Antenna pointing error exceeding 1.0 degrees for more than 60 seconds	Caution	Warning
S-31	**Communications**	Received signal strength declining 15 dB below nominal without geometry explanation	Watch	Warning
S-32	**Communications**	Transmitter output power dropping 20% below rated	Watch	Caution
S-33	**Communications**	Data uplink rate collapsing to below 10% of normal	Caution	Warning
S-34	**Communications**	Antenna gimbal motor current spiking during pointing maneuvers	Watch	Caution
S-35	**Thermal + Power**	Thermal runaway in battery pack (correlated thermal and power fault)	Warning	**Emergency**
S-36	**Power + Life Support**	Life support power feed voltage sagging while CO2 is rising (related faults)	Caution	**Emergency**
S-37	**Navigation + Propulsion**	Attitude error triggering unexpected thruster firing (cascade fault)	Watch	Warning
S-38	**Thermal + Life Support**	Life support heat exchanger temperature anomaly affecting cabin temperature	Watch	Caution
S-39	**All Systems**	Gradual power reduction affecting all systems (solar array degradation)	Watch	Warning
S-40	**Thermal**	Heater element open-circuit failure (reads zero output)	Caution	Warning
S-41	**Power**	Battery internal resistance rising over 48 hours (aging indicator)	Watch	Caution
S-42	**Life Support**	Potable water system pressure anomaly (secondary life support)	Watch	Watch
S-43	**Navigation**	Star tracker temporarily blinded by Earth or Moon limb (expected transient)	Watch	Watch
S-44	**Propulsion**	Propellant pressurant leak reducing tank pressure over 12 hours	Watch	Warning
S-45	**Communications**	Antenna heat shield thermal anomaly affecting pointing accuracy	Watch	Caution
S-46	**Thermal**	Multi-layer insulation degradation detected via increasing thermal loss rate	Watch	Caution
S-47	**Power**	Inverter frequency anomaly on AC power bus	Caution	Warning
S-48	**Life Support**	Trace contaminant level rising in cabin air (slow chemical off-gassing)	Watch	Caution
S-49	**Navigation**	Momentum management system requiring more frequent desaturation	Watch	Watch
S-50	**Propulsion + Power**	Low-thrust propulsion test consuming more power than predicted	Watch	Caution

5. Level 4: Human Expert Review

5.1 Purpose

Automated testing confirms that the system does what it was programmed to do. It cannot confirm that what it was programmed to do is actually the right thing. Human expert review addresses that gap by having reviewers who were not involved in designing the system evaluate its decisions against their own professional judgment.

5.2 Review Panel

Reviewer Role	Count	Qualifications
Faculty advisor	1	Aerospace engineering faculty member with experience in spacecraft fault management systems
Senior engineering students	5	Students in aerospace or mechanical engineering with coursework in spacecraft systems
Independent reviewer	1	Individual with no prior knowledge of DeepGuard's design or intended responses

5.3 Review Process

- **Scenario selection.** 20 of the 50 Level 3 scenarios are selected for human review, including all four alert levels, all six spacecraft systems, and at least 3 multi-system fault scenarios.
- **Blind review.** Reviewers receive the sensor readings and the situation description but not DeepGuard's response. They independently write down what alert level they would assign and what action they would recommend.
- **Comparison.** Reviewer recommendations are compared to DeepGuard's actual decisions and scored as: exact match (full agreement), one level off (partial agreement), or two or more levels off (disagreement).
- **Discussion of disagreements.** Any scenario where fewer than 4 of 7 reviewers agree with DeepGuard's decision is discussed as a panel. The panel decides whether the system's logic should be revised before the design is finalized.

> **Pass condition for Level 4:** At least 18 of the 20 reviewed decisions receive full or partial agreement from at least 5 of the 7 reviewers. Any decision receiving full disagreement from 4 or more reviewers is flagged for design revision before the system is considered review-complete.

6. Evaluation Metrics in Plain English

Every measurement used to evaluate DeepGuard is defined below in plain language, without technical shorthand.

Metric Name	What It Actually Measures	How It Is Calculated	Target	Why This Target
Problem detection rate	Out of every 100 real faults, how many does the system correctly flag?	Real faults correctly flagged divided by total real faults tested, times 100	> 87%	NASA crewed spacecraft standard; below 87% is considered too unreliable for crew safety

Metric Name	What It Actually Measures	How It Is Calculated	Target	Why This Target
False alarm rate	Out of every 100 normal readings, how many does the system wrongly flag as suspicious?	Normal readings incorrectly flagged divided by total normal readings tested, times 100	< 5%	Above 5%, the crew begins to ignore alerts; this is documented in aviation and spaceflight safety research
Detection and alarm balance	A single score rewarding high detection and low false alarms at the same time; falls between 0 and 1	(2 x detection rate x (1 minus false alarm rate)) divided by (detection rate plus (1 minus false alarm rate))	> 0.86	A score below 0.86 shows an unacceptable trade-off between catching faults and avoiding false alarms
Emergency response time	How long from a sensor reading arriving to the emergency response protocol starting	Milliseconds from sensor packet arrival to first response command, measured at the 99th percentile	< 2 sec	99th percentile ensures even the slowest responses meet the target, not just the average case
Forecast accuracy	How close the 15-minute predictions are to what the sensors actually read at that future time	Average of the absolute difference between predicted and actual values, as a percent of each sensor's full range	< 2%	A 2% error on a 200-unit sensor equals 4 units; small enough to support accurate alert level decisions
Human agreement rate	What percentage of DeepGuard's decisions do aerospace engineering experts agree with?	Decisions receiving at least partial agreement from 5 of 7 reviewers, divided by total decisions reviewed, times 100	> 90%	Establishes that the system's logic aligns with human professional judgment, not just internal test criteria
Emergency fault miss rate	How many Emergency-level faults does the system fail to detect across all simulation runs?	Count of Emergency-level scenarios where no Emergency alert was generated across all 500 simulation runs	Zero	No Emergency-level fault can ever be missed; any missed Emergency is an unacceptable safety failure

7. Documentation Standards

7.1 Audit Log Field Specification

Every automated action DeepGuard takes produces one audit log entry written to a tamper-resistant binary log file within 10 milliseconds. The log file is checksummed every 60 seconds to detect any corruption from radiation events.

Field	Format	Example	Purpose
Timestamp	UTC date and time to the nearest millisecond	2026-05-05T14:23:07.412Z	Allows engineers to cross-reference with the mission timeline
Sensor ID	System code plus sensor number	ECLSS-CO2-01	Identifies exactly which sensor triggered the event
Raw sensor value	Number in engineering units	0.78 kPa	Records the actual reading that caused the alert
Suspicion score	Negative decimal from the Sensor Watcher	-0.22	The score that triggered the alert flag
Forecast value	Predicted value plus time horizon	0.91 kPa in 15 min	The Trend Forecaster's prediction that influenced severity classification
Alert level	Text label	CAUTION	The level the Action Engine assigned to this event
Protocol ID	Text code	L-01	The specific approved response protocol that was run
Action description	Plain-English text	Increased CO2 scrubber cycle rate by 25%	Human-readable record of what the system did
Crew notified	Yes or No	Yes	Confirms whether a crew alert was generated for this event
Checksum	32-character verification string	a3f7c... (32 characters)	Verifies the log entry has not been corrupted by radiation or hardware error

7.2 Document Library

Document	File Name	Contents	Graded Section
Project Proposal	FP_Robert_B_Thompson_ITAI2372_Proposal.pdf	Problem statement, solution overview, objectives, data sources, timeline, and references	Proposal (20 pts)
Main Development and Design	FP_Robert_B_Thompson_ITAI2372_MainDesign.pdf	Full system architecture, detailed design of all three AI tools, hardware specs, risk analysis, ethical boundaries	Main Design (40 pts)
Documentation and Testing	FP_Robert_B_Thompson_ITAI2372_DocTesting.pdf	All four testing levels, 50 fault scenarios, evaluation metrics, audit log spec, maintenance plan	Doc and Testing (25 pts)

Document	File Name	Contents	Graded Section
Presentation Slides	FP_Robert_B_Thompson_ITAI2372_Slides.pdf	12-slide visual summary of the full project for the final presentation	Presentation (15 pts)

8. Long-Term Maintenance Plan

A machine-learning system that is not updated will become less accurate as conditions change. The table below defines the specific activities that keep DeepGuard accurate across a multi-year deep space mission.

Activity	Frequency	Who Does It	What Triggers an Alert or Rollback
Sensor Watcher re-learning	Every 30 mission days	Runs automatically on SpaceCube-3	If detection rate on a validation set falls below 85% after re-learning, the previous model version is restored and a crew alert is sent
Trend Forecaster re-training	Every 90 mission days	Runs automatically overnight during low-activity period	If average prediction error rises above 3% after re-training, the previous model is restored and a crew alert is sent
Audit log integrity check	Every 60 seconds	Runs automatically; no crew involvement needed	If any checksum fails, the affected log segment is flagged and ground control is notified at the next communication window
Protocol library review	Before each major mission phase	Mission engineers on the ground	Any protocol added or removed requires a software update transmitted from Earth and confirmed by the crew before taking effect
Full system performance review	Every 6 months or after any Emergency event	Ground mission engineering team	If any metric falls below its target threshold, ground control issues a maintenance procedure for the crew to follow
Emergency model rollback	As needed, within 10 minutes	Automatic if triggered; crew-confirmed if manual	If both the current model and the re-learned model fail performance checks, the factory-default model shipped at launch is restored immediately

9. References

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